Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/09/2004
PI Name:	Balakotaiah, Vemuri Ph.D.		
Project Title:	Fundamental Studies on Two-Phase Gas-Liquid Flows through Packed-Beds in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77204	**Congressional District:**	18
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	01/17/2002	**End Date:**	10/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	McCready, Mark (University of Notre Dame) Motil, Briam (NASA Glenn Research Center)		
Grant/Contract No.:	NAG3-2697		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In the typical operation of a packed-bed reactor, gas and liquid flow simultaneously through a fixed bed of solid particles. Depending on the application, the particles can be of various shapes and sizes and provide for intimate contact and high rates of transport between the phases needed to sustain chemical or biological reactions. The packing may also serve as either a catalyst or as a support for growing biological material. NASA has flown two of these packed-bed systems in a microgravity environment with limited or no success. The goal of this research is to develop models (with scale-up capability) needed for the design of the physicochemical equipment to carry out these unit operations in microgravity. New insight will also lead to improvements in normal gravity operations. The main objective of this task is to provide a quantitative prediction based on analysis and verified by experiments of gas-liquid flow in packed-beds. It will provide a fundamental understanding of cocurrent two-phase flow in packed-beds in microgravity. The information and modeling from this task will significantly contribute to the fundamental understanding of the interactions between the surface tension, inertia and viscous forces in the absence of gravity for packed bed reactors. The packed-bed reactor is a critical unit operation for a wide variety of physicochemical and biological technologies including regenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This task supported NASA’s shift to strategic-based research during most of the year. Gas-liquid flow through fixed bed reactors is a critical component for many water processing and air regeneration systems. The Principal and Co-Investigators worked with NASA to shift the focus of these studies to several well defined technologies. Much of the effort involved defining requirements for an ISS experiment. The space flight experiment will verify hydrodynamic models for reactor configurations similar to designs planned for water reclamation systems. A Requirements Definition Review (RDR) is scheduled for March 2005. A new test facility for the reduced gravity aircraft was completed and low gravity tests were conducted over the course of two flight weeks using a new larger scale cylindrical test section with different packing materials. A void fraction sensor developed by Creare, Inc. under an SBIR was also tested during these flights. A preliminary review of the flight data has confirmed all diagnostics functioned very well. Analysis of the flight data will be completed in the first quarter of FY05 to support the RDR.
Bibliography Type:	Description: (Last Updated: 01/12/2016)
NASA Technical Documents	Balakotaiah V, Motil BJ, McCready MJ. "Operating Packed Bed Reactors in Space." NASA Technical Document, May 2004. , May-2004